Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp. 1-4 **Animal Science**
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Ameliorative role of ascorbic acid (vitamin C) against hydrogen peroxide toxicity on protein level in brain and testis of fish *Catla catla* (Ham.)

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Received : 11th April, 2025 ; Revised : 10th May, 2025
DOI:-https://doi.org/10.5281/zenodo.18587061

Abstract- For this experiment fish *Catla catla* has been chosen as the model animal. The whole experiment was conducted around the month May to September in 2024 during the breeding period of the fish *Catla catla*. Fishes were collected from different sources. The whole experiment was started with around 100 fishes having a weight of about 90±10 g. Few fishes died in the starting period during acclimatization. Around 60 fishes survived for the dosing process. All the rest fishes were then divided into four groups having 15 fishes in each group and dosing was initiated. In the first group animals were provided only fish diet and the group was labelled as control group. In second group Hydrogen peroxide (15µl/l) was added and this group was labelled as Group-II. In the third group Vitamin-C (20mg/l) along with Hydrogen Peroxide (15µl/l) was added and this group was labelled as Group-III. In the last fourth group only Vitamin-C (20mg/l) was added and this group was labelled as Group-IV. The dosing process continued for the time period of 30, 60 and 90 days. After the completion of dosing, animals were sacrificed and required tissue were collected for the further analysis and collection of biochemical experimental data for getting the protein content value. After observing and analyzing the collected data it was concluded that the amount of total protein in the tissues of brain and testis was insignificantly increased in the group that was treated with Hydrogen peroxide only while this increase was lesser in the group that was treated with hydrogen peroxide together with Vitamin-C and minimal in the group treated with only vitamin-C as compared to the Control group. All these result shows that hydrogen peroxide have some toxic effects which got reduced by the antioxidant properties of vitamin-C in fish *Catla catla.*

Keywords: Protein, hydrogen peroxide, vitamin-C, *Catla catla*

INTRODUCTION

Hydrogen Peroxide (H_2O_2) toxicity for fish has attracted much attention, and it has varied considerably in different fish species, life stages, and temperatures. For instance, the 1 h-LC_{50} (median lethal concentration) value of H_2O_2 is 1640–2480 ppm in channel catfish (*Ictalurus punctatus*) and 183–260 ppm in rainbow trout (*Oncorhynchus mykiss*) at 22°C.[1]

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The tolerance concentration was less than 0.335 mM in blue gourami (*Trichogaster trichopterus*) and 0.194 mM in suckermouth catfish (*Hypostomus plecostomus*) for 1h of H_2O_2 exposure.[2]

Hydrogen Peroxide is the simplest peroxide having the formula H_2O_2. Its pure form is a pale blue liquid, slightly more viscous than water.[3] H_2O_2 is used around the world in a variety of commercial, industrial, medical, environmental, and personal hygiene applications.

H_2O_2 can be toxic by different routes like ingestion, inhalation, skin contact and eye exposure.[4] H_2O_2 exposure

also caused oxidative stress, leading to damage of the intestinal epithelial cells in Jian carp (*Cyprinus carpio* var. Jian)[5] and the liver in tilapia (*Oreochromis niloticus*).[4]

Vitamin-C (Ascorbic acid) is a water-soluble vitamin it is necessary for growth and repair of tissues in all parts of the body. Vitamin-C also helps in signal transduction and is also involved in several physiological processes, such as immune stimulation, synthesis of collagen, hormones, neurotransmitters, and iron absorption, has also roles in detoxifying the body of heavy metals.[5]

Vitamin-C has important role in a great number of biochemical processes such as collagen which is an intercellular protein and principal constituent of skin, scales, mucosa, cartilaginous tissues, bones and conjunctive tissue formation, which involves all the organs of the body.[6] Vitamin C (Ascorbic acid) is also one of the micro nutrient elements in feeding teleost fish, ascorbic acid plays a critical role in all living organisms. Trout, salmon and other fish species have dietary requirements for Ascorbic acid.[7]

The aim of this study is to analyse the toxic impacts of hydrogen peroxide and role of vitamin-C as an antioxidant by calculating the total protein content in the tissues of brain and testis.

MATERIALS & METHODS

The model animal of this experiment i.e., fish *Catla catla* were collected from different available sources. The whole experiment was conducted around the month May to September 2024 during the breeding period of the fish *Catla catla* specially for collecting the gonads tissue. The experiment was started with around 100 fishes having a weight of about 90±10 g. All the fishes then kept in suitable laboratory environment in preferable size of aquarium for the proper acclimatization during this period of time, fish were provided only fish diet. The water quality and fishes' survival checked frequently as in the starting period after keeping them in laboratory environment their mortality rate is highest, they gradually get adopted in the laboratory environment and once they get adopted their mortality rate get decreased too minimal. Among the 100 fishes few fishes died in the starting period during the acclimatization process. Around 80 fishes survived for the dosing process. After completion of acclimatization all the rest fishes were then divided into four groups with 15 each and dosing was initiated. In the first group, animals were provided only fish diet and this group were labelled as the Control group. In the second group 15µl/l Hydrogen peroxide was added and this group was labelled as Group-II. In the third group 20mg/l Vitamin-C along with Hydrogen Peroxide was added and this group was labelled as Group-III. And in the last fourth group only Vitamin-C was added and this group was labelled as Group-IV. The dosing process continued for the time period of 30, 60 and 90 days. After the completion of dosing animals were sacrificed on 31st,61st and 91st day and required tissue i.e., brain and testis were dissected out and weighed by using digital weighing machine. After weighing, the tissues were homogenated by using distilled water then after tissues were centrifuged for the required duration of time. And lastly after the completion of centrifugation Total Protein content, was estimated by using the McDowell method.[8]

RESULT

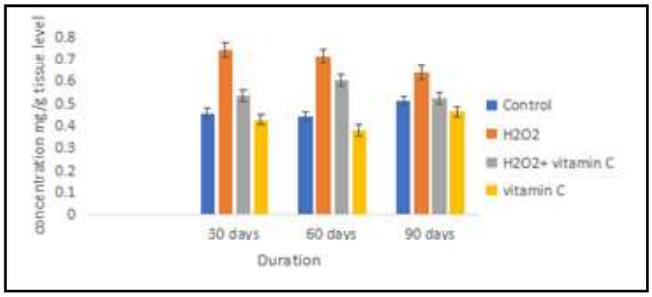
Fishes *Catla catla* were exposed with H_2O_2 showed change in color little red. Treatment Hydrogen Peroxide (H_2O_2) (15µl/l) and vitamin-C (20µl/l) along with Hydrogen Peroxide (H_2O_2), and vitamin-C alone showed recovery in the fishes. When the fish *Catla catla* exposed with H_2O_2 showed insignificant changes in total protein level in brain and testis after 30, 60 and 90 days as compared to control. While this increase was lesser in the group treated with vitamin-C along with hydrogen peroxide (H_2O_2) group as compared to H_2O_2 group, and in group treated with vitamin-C alone showed lesser in Brain in comparison to the H_2O_2 group, whereas, in testis it showed insignificantly lesser than Control group in *Catla catla*. All these result shows that hydrogen peroxide have some toxic effects which got reduced by the antioxidant properties of vitamin-C.

Table 1. Protein level mg/g concentration in Brain of *Catla catla* of H_2O_2, H_2O_2 along with vitamin C and vitamin C exposures.

Duration	30 days	60 days	90 days
Control	0.457±0.045	0.441±0.035	0.512±0.048
H_2O_2	0.742±0.075	0.715±0.045	0.637±0.050
H_2O_2+vitamin C	0.535±0.039	0.605±0.041	0.526±0.034
Vitamin C	0.429±0.036	0.379±0.025	0.464±0.035

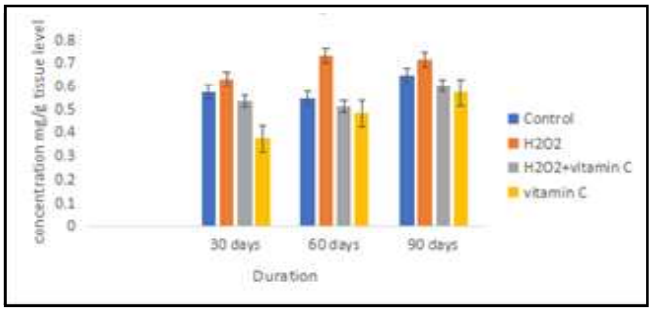
Table 2. Protein level mg/g concentration in Testis of *Catla catla* of H_2O_2, H_2O_2 along with vitamin C and vitamin C exposures.

Duration	30 days	60 days	90 days
Control	0.576±0.083	0.544±0.090	0.648±0.058
H_2O_2	0.633±0.053	0.731±0.071	0.717±0.054
H_2O_2+vitamin C	0.535±0.039	0.512±0.057	0.603±0.044
Vitamin C	0.377±0.048	0.483±0.049	0.572±0.070



Values are mean ± SEM of 5 fishes

Fig. 1- Protein level mg/g concentration in Brain of *Catla catla* of H_2O_2, H_2O_2 along with vitamin C and vitamin C exposures.



Values are mean ± SEM of 5 fishes

Fig. 2- Protein level mg/g concentration in Testis of *Catla catla* of H_2O_2, H_2O_2 along with vitamin C and vitamin C exposures.

DISCUSSION

The toxicity of Hydrogen Peroxide (H_2O_2), largely mediated by hydroxyl radicals (OH) generated by the Fe^{2+} -catalyzed Fenton reaction[9] may inflict alteration to brain molecules. Brain was most vulnerable to oxidation due to limited antioxidant system[10] that in turn, the main target for drug effect and adverse effects because it contains the highest number of receptors[11]. The brain is believed to be particularly vulnerable to the damaging effect of H_2O_2.[12]

Hydrogen Peroxide (H_2O_2) exposure induced brain injury in fish *Cyprinus carpio*.[4] Hydrogen Peroxide (H_2O_2) can induce oxidative stress in fish, damaging proteins and other biomolecules in the brain and testis.[4] H_2O_2 when comes to brain and testis tissues, exposures to H_2O_2 can lead to significant protein level alterations. Thymus algeriensis (TEO) protect testis from H_2O_2 generated oxidative stress in Rat.[13] In this study, it showed that Protein level were increased insignificantly in H_2O_2 group in 30, 60 and 90 days as compared to control group.

These results are much expected because ascorbic acid (Vitamin-C) is closely related to the immunological system performance, and has antioxidant properties, favors integrity and fluidity of membranes[14] controlling the oxidizing reactions of fatty acids thus keeping cellular respiration and avoiding cell death.[15] This antioxidant activity of ascorbic acid (Vitamin C) makes it a hunter of free radicals, thus preventing the autointoxication of immunological cells such as macrophages which are the first processors of the information about the alien bodies and maximizing the defense of fish.[14] In fish the ascorbic acid concentrations were significantly observed in the testis.[16]

Brain requires molecular oxygen in large quantities, which is then followed by the formation of free radicals in high levels. The brain also contains polyunsaturated fatty acids and oxidized easily.[17,18] In addition, the total capacity of antioxidants found in the central nervous system are relatively few.[17,19]

Excessive levels of free radicals will damage many cellular components, including cellular proteins, DNA, phospholipids membrane and enzyme inactivation.[17] It also causes a loss of function of specific proteins, clearance of abnormal proteins, cellular redox balance depletion and disruption of the cell cycle, and ultimately cause death of nerve.[17,18] Dietary vitamin C supplementation might be helpful in abrogation of chlorpyrifos toxicity and improves growth survival and biochemical test in brain tissue of *Clarias batrachus.*

CONCLUSION

In this investigation, it showed that supplementation of vitamin-C along with H_2O_2 insignificantly lowered protein level in 30, 60 and 90 days as compared to H_2O_2 group. While, supplementation of vitamin-C alone showed lesser protein level in Brain and testis of *Catla catla*. In this work, Vitamin-C acting as an antioxidant and can mitigates these harmful effects by neutralizing free radicals and protecting against protein damage.

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